



Design and Implementation of a Real-Time

Cartesian Impedance Controller for a 7-DOF Collaborative Robot

Contact-induced tool-surface alignment with a virtual centre of compliance

Heykel Khadhraoui

Master Thesis Defense · Computational Engineering (M.Sc.)

Supervisors: Prof. Norbert Nitzsche · Prof. Ruth Otto

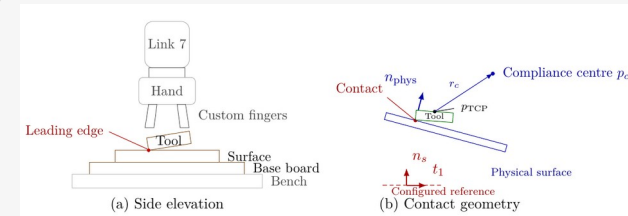


Figure 4.1: Experimental system: (a) physical set-up and (b) surface-frame contact geometry.

Robot, tool and surface-contact geometry

Presentation overview

From the contact problem to the measured controller behaviour

1

Motivation and objective

2

Controller and CoC theory

3

Implementation and experiments

4

Results and interpretation

5

Conclusion and outlook

Main question

How do
stiffness and
virtual CoC
position
influence
passive tool-
surface
alignment
during Contact
Establishment?

Motivation

Angular mismatch at contact entry can create edge-first loading

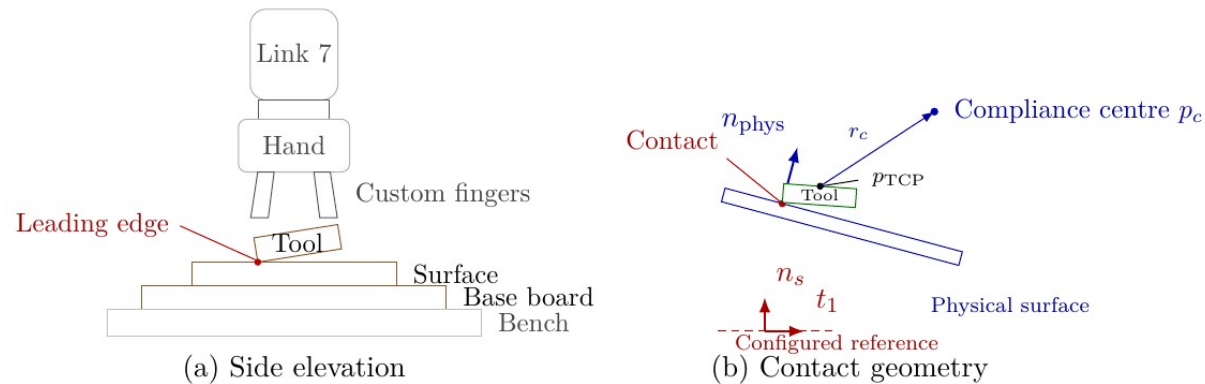


Figure 4.1: Experimental system: (a) physical set-up and (b) surface-frame contact geometry.

Surface placement, calibration and tool mounting introduce small orientation differences before contact.

A tilted rectangular face can first contact along an edge or corner, producing concentrated contact loads.

Rigid position control resists the corrective rotation. Finite rotational compliance allows contact interaction to rotate the tool towards the surface.

The aim is to keep the normal pressing action while allowing passive contact-induced alignment.

Experimental tool and surface geometry used in the thesis.

Thesis objective and contributions

What was implemented and what was measured

Controller contribution

Real-time Cartesian impedance controller for the Franka Emika Panda.

Directional translational and rotational stiffness and damping defined relative to the surface.

Software-defined virtual centre of compliance displaced from the TCP.

Projected null-space damping and singular-value conditioning.

Experimental contribution

Main contact study: Cases A–D.

Case A: TCP-centred baseline.

Case B: rotational stiffness.

Case C: translational stiffness along the surface.

Case D: signed tangential CoC position.

Separate Cartesian pose-hold study for the null-space controller.

Cartesian impedance controller

The controller regulates position and orientation directly in Cartesian space

Impedance wrench

$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE})$$

$$m = K_R e_R + D_R (\omega_d - \omega_{EE})$$

The surface frame provides the physically relevant directions: t_1 , t_2 and n_s . Stiffness and damping are assigned separately along/about these directions.

Joint-torque command

$$\tau_{cmd} = J^T F + \tau_{null} + \tau_c(q, \dot{q})$$

J^T maps the commanded Cartesian wrench to joint torque.

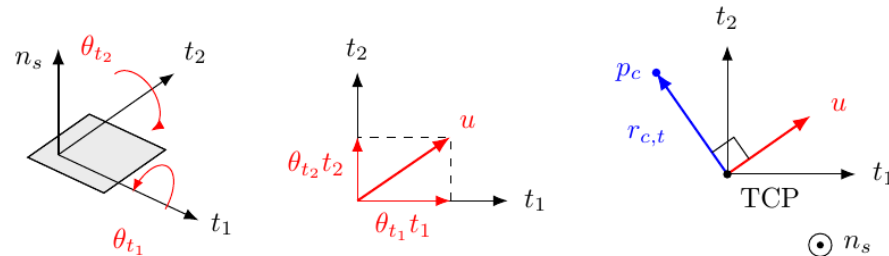
τ_{null} is an additive secondary torque projected into the kinematic null space.

τ_c supplies the Coriolis/centrifugal model term required by the robot interface.

Virtual centre of compliance

A displaced reference couples normal pressing to a selectable rotational moment

... and the displacement selected perpendicular to that direction.



(a) Tangent rotations (b) First-order direction (c) Selected displacement

Figure 2.3: Tangent rotations, their combined direction, and the selected displacement

Small-angle tilt direction and the selected tangential CoC displacement.

$$m = m_R + r_c \times f$$

$$u = \frac{\theta_{t_1} t_1 + \theta_{t_2} t_2}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}}$$

$$r_{c,t} = \|r_{c,t}\| (n_s \times u)$$

$$r_{c,t} \times f_n = \|r_{c,t}\| F_n u, \quad F_n < 0$$

Interpretation

The CoC magnitude sets the lever length, while its tangential direction sets the sign and axis of the additional TCP moment generated by the press.

Real-time implementation

One 1 kHz callback evaluates the reference, impedance wrench and torque command

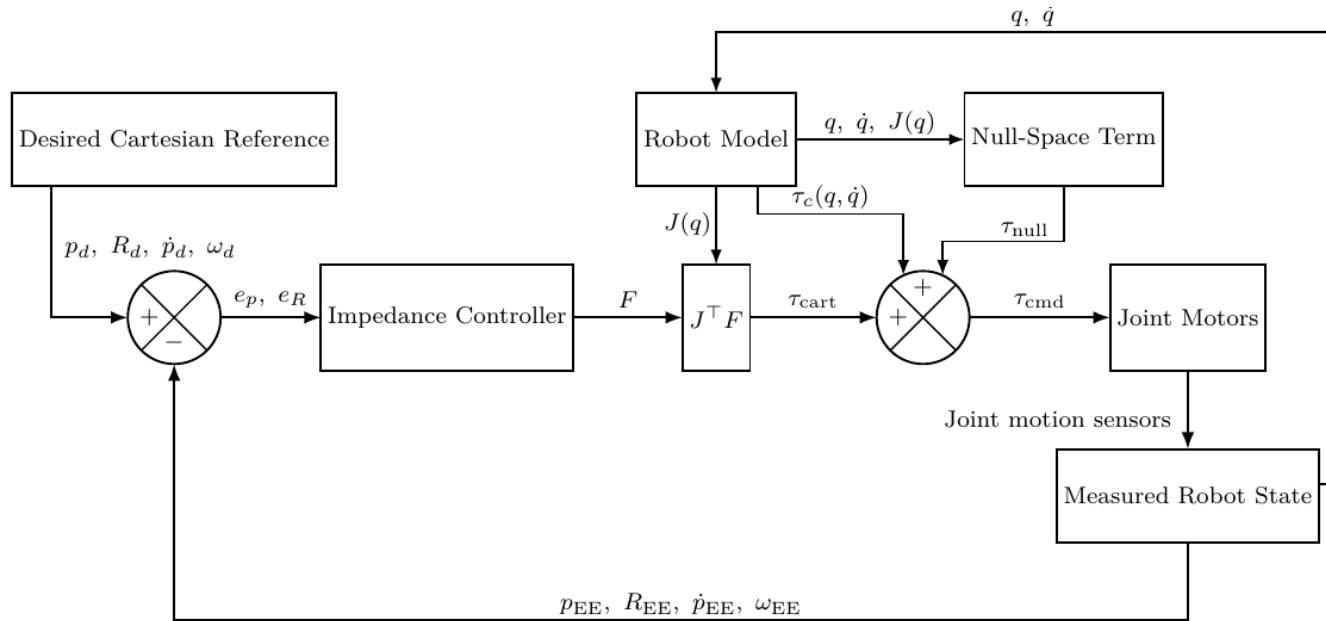


Figure 3.2: Cartesian impedance feedback loop and additive torque terms.

Cartesian impedance feedback loop and additive torque terms.

The active state provides the desired Cartesian reference and the state-specific gains.

The measured robot state supplies pose, velocity, joint state and Jacobian information.

During Contact Establishment, the orientation reference is held at the entry orientation; the measured rotation is therefore contact-induced.

The same torque-level architecture is used for the contact study and the separate pose-hold study.

Contact sequence and experiment design

The measured response is generated during Contact Establishment

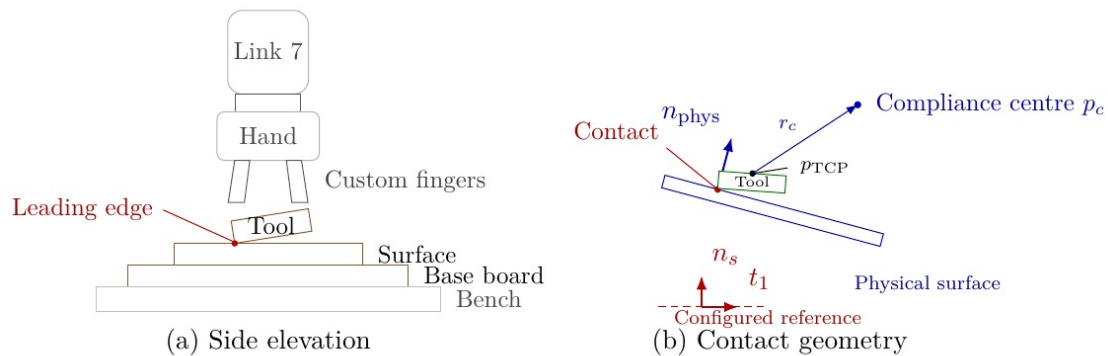


Figure 4.1: Experimental system: (a) physical set-up and (b) surface-frame contact geometry.

Main contact matrix

Case A — CoC at TCP, positive/negative/zero offset.

Case B — rotational stiffness about t_1 .

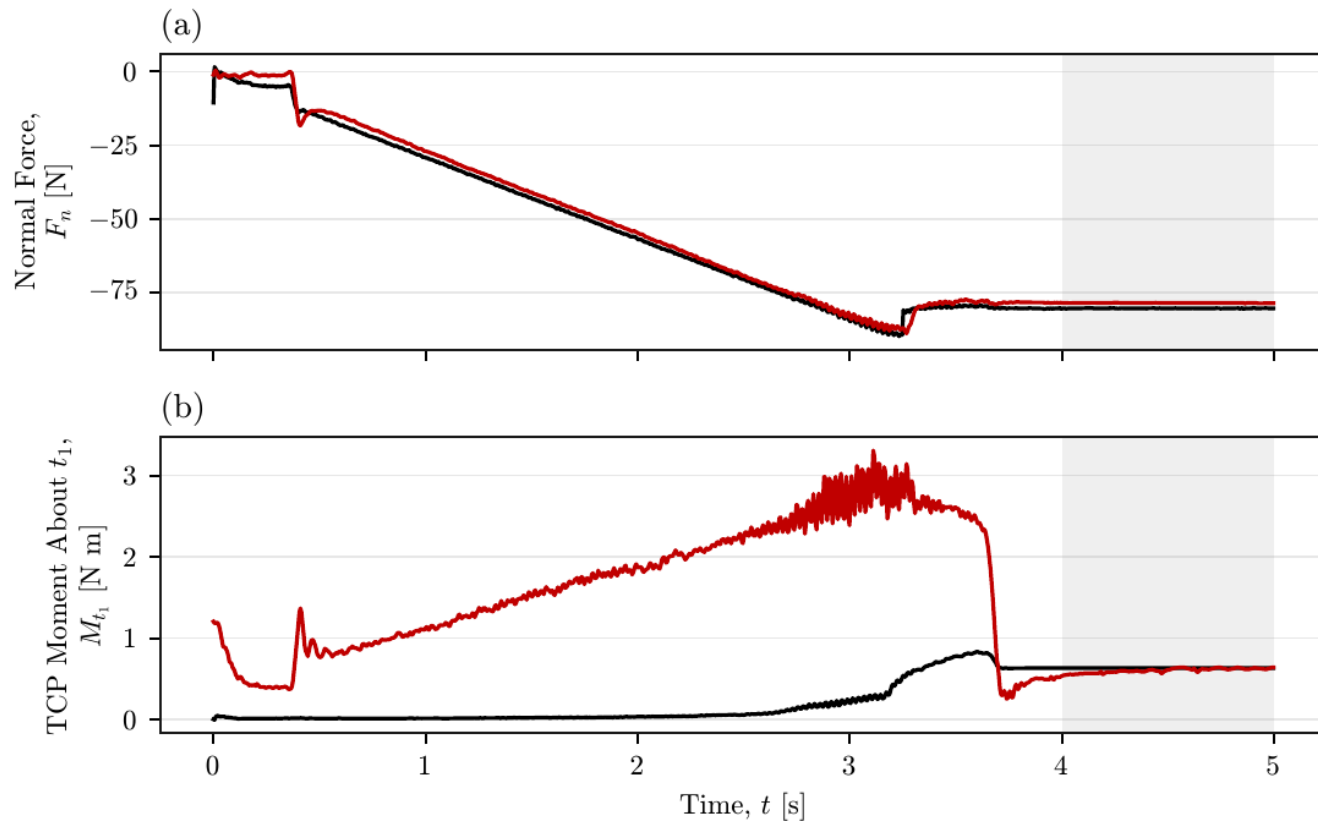
Case C — tangential translational stiffness along t_2 .

Case D — tangential CoC position for both initial offset directions.

Three repetitions per parameter setting.

Controller wrench evaluation

Commanded and model-estimated interaction quantities agree closely in stationary contact



2.32%

normal-force difference

4.87%

moment difference

The comparison uses one TCP-centred positive-offset Contact Establishment trial.

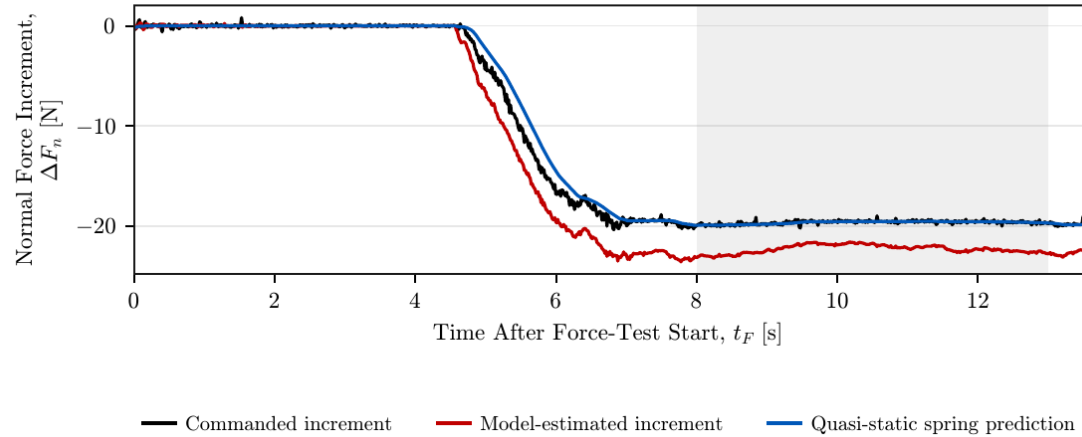
Both commanded and model-estimated quantities are expressed at the TCP and resolved on the same surface axes.

The small stationary differences support using the robot's model-estimated wrench as the measured interaction wrench available from the robot.

Quasi-static plausibility assessment

Separate translational and rotational perturbations check the implemented stiffness relations

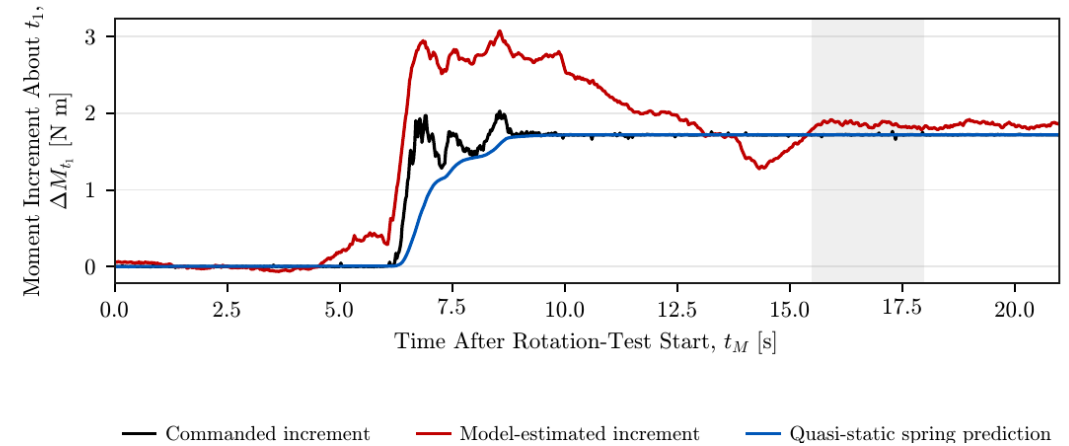
Chapter 5. Results and Discussion



Normal-force perturbation

$$F_{n,qs} = K_{p,n} n_s^T (e_p - e_{p,0})$$

Commanded moment agreed to the reported precision, while the model estimated moment differed from the command by 7.75 %. This comparison supports the plausibility of the implemented rotational impedance relation.



Moment perturbation about t_1

Quasi-static prediction vs commanded: 0.05% for force and agreement to reported precision for moment. Model-estimated differences were 13.50% and 0.75%, respectively.

$$M_{t_1,qs} = K_{p,t_1} t_1^T (e_p - e_{p,0})$$

Main results I — baseline and stiffness

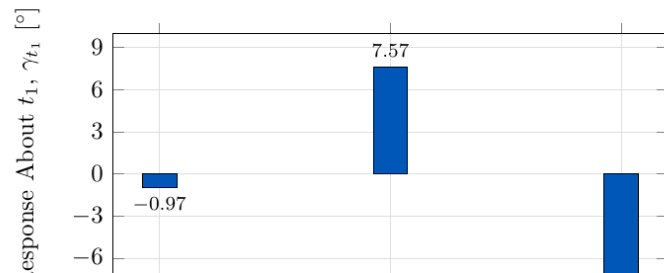
Cases A–C establish how strongly the response depends on the basic impedance settings

0° , 10° , and -10° . These conditions establish the TCP-centred stiffness and CoC-position comparisons.

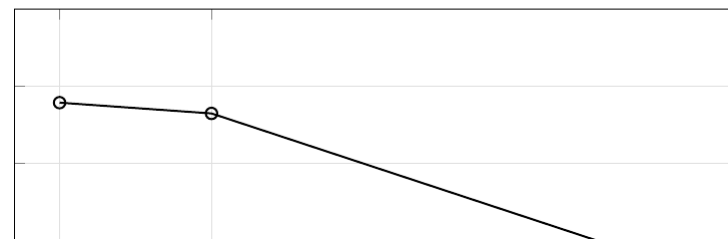
For the zero-offset condition, the initial condition is reported by the attitude ϕ_0 . For the 10° and -10° conditions, the achieved response is reported by θ_{init,t_1} .

Case B: how the rotational stiffness about t_1 affects the correlation. The experiment was performed with the positive 10° configured orientation. The response was compared at 5, 15, and 50 N m/rad. The 5 N m/rad condition is the baseline from Case A, while the remaining controller settings

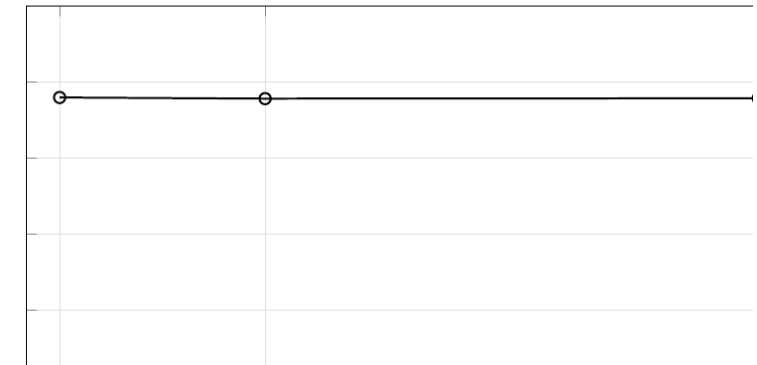
were kept at $K_{R,t_1} = 5$ N m/rad, while the translational stiffness was kept at $K_{p,t_2} = 2000$ N/m. The $K_{p,t_2} = 2000$ N/m condition is the reference from Case A.



Case A — TCP baseline



Case B — rotational stiffness



Case C — translational stiffness

Case A: contact-induced rotation occurred towards the configured surface for both tested initial rotational directions.

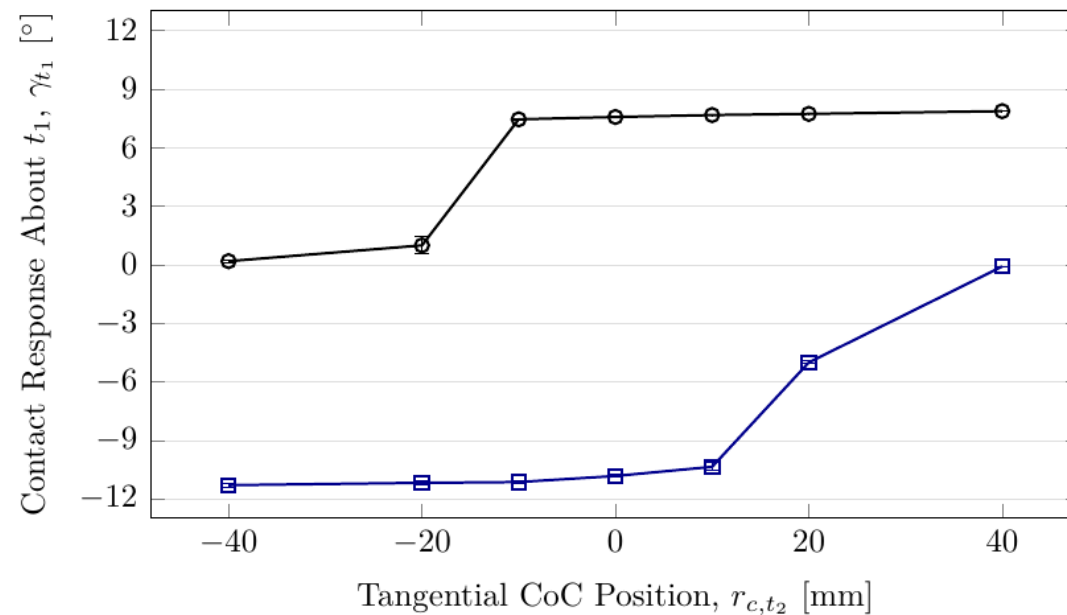
Case B: increasing K_{R,t_1} from 5 to 50 N m/rad reduced the response from 7.57° to 2.73° — a 63.9% reduction.

Case C: the tested K_{p,t_2} variation changed the response by only 0.03° , equal to about 0.4% of the TCP-centred reference.

Main results II — tangential CoC position

The CoC displacement strongly changes the preferred rotational direction

mm, where $r_{c,t_2} = 0$ is the TCP-centred reference from Case A. Cartesian st
ng, and the remaining controller settings were kept unchanged.



Achieved Initial Angular Offset, $\theta_{init,t_1} = 9.32^\circ$ —■— Achieved Initial Angular Offset, $\theta_{init,t_1} =$

Figure 5.7: Contact response by tangential CoC position.

Case D findings

For the positive offset, moving the CoC from -40 mm to $+40$ mm increased the measured response from 0.18° to 7.87° .

For the negative offset, the same CoC sweep changed the response from -11.30° to -0.09° .

A suitably selected displaced CoC increased the final response only slightly ($\sim 4\%$) but produced faster alignment.

The opposite displacement direction reduced the response by approximately 98–99%.

Main interpretation: the displaced CoC acts primarily as a selector of the preferred rotational direction.

Main results III — mechanism behind the CoC effect

The estimated contact moment changes strongly while the normal force remains similar

Chapter 5. Results and Discussion

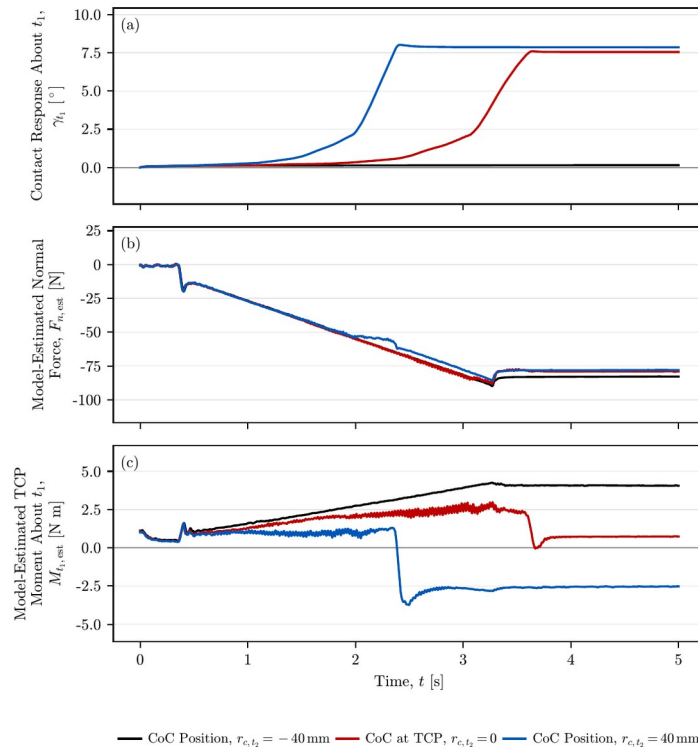


Figure 5.8: Contact response and model-estimated interaction wrench for the three CoC positions.

Figure 5.8 compares the contact response γ_{t_1} in panel (a), the model-estimated normal force $F_{n,est}$ in panel (b), and the model-estimated TCP moment about the first surface tangent $M_{t1,est}$ in panel (c), on a common time axis for the three centre positions. Over the

Final stationary values

CoC position: -40 mm TCP
 $+40$ mm

Estimated F_n [N]
 -82.9 -79.0 -78.0

Estimated M_{t1} [N m]
 $+4.07$ $+0.73$ -2.54

The normal force changes little, whereas the TCP moment changes sign and magnitude. The measured rotational response follows this change.

$$m = m_R + r_C \times f$$

Secondary study — null-space pose hold

Projected damping and singular-value conditioning modify redundant joint motion while the Cartesian pose is held

Chapter 5. Results and Discussion

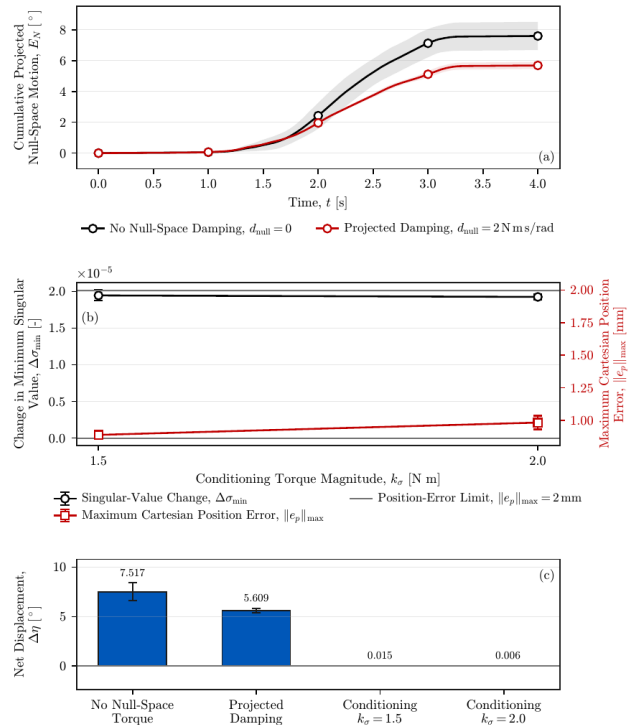


Figure 5.9: Null-space pose-hold response.

Figure 5.9 summarises the Cartesian pose-hold disturbance test in three panels. Panel (a) shows the mean cumulative projected null-space motion E_N for the inactive and projected-damping modes, with the shaded bands indicating the spread across the repeated trials. Panel (b) shows the singular-value change $\Delta\sigma_{min}$ together with the maximum Cartesian

Measured effects

Projected damping reduced cumulative projected null-space motion from 7.596° to 5.687° — about 25%.

Singular-value conditioning reduced the net displacement from 7.517° to 0.015° and 0.006° for the two tested torque magnitudes.

This corresponds to approximately 99.8–99.9% reduction in net displacement.

All tested conditioning cases remained below the 2 mm Cartesian position-error criterion.

$$N_\tau = N_q^\top$$

The larger conditioning torque produced more cumulative back-and-forth motion despite a similar final region.

Main findings

What the experiments establish together

- 1 Rotational compliance enables passive contact-induced alignment.
- 2 Rotational stiffness has a strong effect; the tested translational stiffness variation has only a small effect.
- 3 **The tangential CoC position strongly biases the rotational response through the additional TCP moment.**
- 4 The TCP-centred CoC is direction-neutral and therefore a sensible final configuration after alignment.
- 5 Projected damping and singular-value conditioning provide complementary null-space behaviour.

Core takeaway: CoC placement is a directional control parameter, not only a scalar lever length.

Limitations and future work

Scope of the present conclusions and the next controller step

Limitations

One robot configuration, one rectangular tool, one configured surface and one Contact Establishment trajectory.

The physical tool-surface angle and instantaneous contact point were not measured directly.

The model-estimated wrench is not an independent force/torque-sensor measurement.

Tool-mount angular play can make the physical tool-face motion differ from the measured end-effector motion.

Grinding itself was implemented but not quantitatively evaluated in this study.

Future work

Measure the physical tool-surface angle and validate the interaction wrench with an external F/T sensor.

Evaluate a more rigid tool mount and additional surface/tool configurations.

Implement an adaptive CoC strategy: start at the TCP, estimate the initial angular offset, temporarily select the supporting tangential displacement, then return the CoC to the TCP after alignment.

Extend the evaluation to sustained grinding and automatic contact-quality measures.

Conclusion

A real-time virtual compliance reference can shape passive tool-surface alignment

The Cartesian impedance controller generated the required normal pressing action while allowing contact-induced rotation towards the configured surface.

The experiments show that the displaced virtual CoC mainly changes the rotational moment and therefore the preferred direction of the alignment response.

A supporting CoC direction can accelerate and slightly increase the response, while the opposite direction can almost suppress it.

This motivates an adaptive CoC strategy that uses a temporary direction-selected displacement and returns to the TCP after alignment.

Thank you

Questions?